Queensland University of Technology
Faculty of Science
School of Mathematical Sciences

Computational Modelling of Multilateral Treaties

Stage 2 — Confirmation of Candidature

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Supervisory Team

Role	Name	Affiliation
Principal Supervisor	Prof. Michael Bode	School of Mathematical Sciences, QUT
Associate Supervisor	Prof. Jonathan Rhodes	School of Mathematical Sciences, QUT

Principal Supervisor — Prof. Michael Bode Prof. Bode is Deputy Director of Capability Development at SAEF (Securing Antarctica's Environmental Future). His work centres on applied mathematics and conservation, with a focus on decision-making under uncertainty and mathematical biology. His previous work on Antarctic Treaty System (ATS) decision-making and conservation makes him an expert on consensus-based systems.

Associate Supervisor — Prof. Jonathan Rhodes Prof. Rhodes is Director of the Centre for Environment and Society. He specialises in spatial ecology, environmental governance, and the modelling of coupled human–natural systems, bridging quantitative models with real-world policy. His expertise gives him deep insight into the policy implications of environmental decisions, providing a grounding for the treaty-system applications explored in this research.

Chapter 1

Background and Literature Review

1.1 Introduction

The Antarctic Treaty System (ATS) is a regime negotiated between a set of parties called consultative parties Secretariat of the Antarctic Treaty (2023). This group of countries, together with a second group lacking decision-making power (non-consultative parties), meets yearly at the Antarctic Treaty Consultative Meeting (ATCM). Before each meeting, culsultative parties may submit Working Papers (WP) and Information Papers (IP) up to 45 days before the ATCM starts Secretariat of the Antarctic Treaty (2023). During ATCMs the parties discuss Working Papers and produce outcomes only by consensus. These outcomes take the form of Measures, Decisions, or Resolutions Antarctic Treaty Consultative Meeting (1995). The consensus requirement makes the ATS a notable example of international cooperation, but is also its main source of fragility. As international relations strain and Antarctic environmental pressures intensify, the risk of losing the capacity to act increases. In 2022, for the first time, the ATCM parties failed to reach consensus on the final report; even as activity has increased, the regime has become less responsive to challenges Gardiner et al. (2025).

In a sistem based on consensus is or utmost importance to uderstand why it succeds or fails, And which parties drive it or block it, respectively. Existing literature has described the ATS by rates and diversity of outputs, and stucturally by describing who co-aouthors with whom Botelho et al. (2026) and Kumar et al. (2026). But there is no study where the a reconstruction of a directional party intention was reconstructed from the textual records on each issue over time. Even further, that type of measure has not been used to model the strategic behaviour that produces or breaks consensus. This thesis objective is to measure that and to build that model and asks if they contain the strategic intentions made by the consultative parties. It then asks whether these can model historical consensus dynamics.

The approach is a computational pipeline. First, it uses embedding techniques to discover topics, and NLI (Natural Language Inference) techniques for stance extraction. These would allow the construction of the preference measure. Second, it applies NLI to extract the opposition or aversion. It then testes the approach by comparing inputs (i.e. working papers) against observes outcomes. Finally, a Multi-Agent reinforcement learning is trained based on the previous outcomes to simulate the negotiation itself.

1.2 The Antarctic Treaty System as a consensus regime

In the ATS, Consultative Parties (CPs) reach outcomes exclusively by consensus. These discussions take place at the annual ATCM, where outcomes can take any of the following forms, per Secretariat of the Antarctic Treaty (2023, Rule 24):

- **Measures** are adopted texts containing provisions intended to become legally binding; these require approval by the relevant parties' legislative bodies.
- **Decisions** are adopted texts concerning internal organisational matters of the ATCM.
- **Resolutions** are non-binding recommendations or expressions of intent.

There are four types of input:

- **Working Papers** (Secretariat of the Antarctic Treaty (2023, Rule 48)): submitted by CPs and Observers, these require discussion at the ATCM and are the main source of outcomes. They are the only input from CPs that can produce an outcome; as such, they are the most likely place for each country to state its intentions.
- **Secretariat Papers** (Secretariat of the Antarctic Treaty (2023, Rule 49)): prepared and presented by the Secretariat, these produce organisational decisions as output, e.g. the budget.
- **Information Papers** (Secretariat of the Antarctic Treaty (2023, Rule 50)): supporting information for Working Papers or other discussions. They may be presented by CPs, Observers, Non-Consultative Parties (NCPs), and experts.
- **Background Papers** (Secretariat of the Antarctic Treaty (2023, Rule 51)): submitted by ATCM participants but not introduced during the meeting; they exist solely for the record.

Of all four, Working Papers are the only input uniquely submitted by CPs, requiring discussion and not confined to organisational matters. They are therefore the most likely place for countries to express their strategic intentions within this closed decision-making environment.

The consensus system makes the ATS a notable example of international cooperation; however, the same mechanism has become a major challenge as the number of CPs grows. Gardiner et al. (2025, Rule 51) showed how the decisio-making responsiveness of the ATS is declining; a clear example of this is the unprecedented 2022 consensus failure.

One good example of consensus efficacy occurred in the late 1970s and 1980s when the negotiation over mineral resources in Antarctica took place. It was a contested episode where the Convention on the Regulation of Antarctic Mineral Resource Activities (CRAMRA) was produced and then abandoned in favour of the Madrid Protocol (1991). This protocol prohibited any mineral resource extraction in the Antarctica (Jackson (2021)). This entertaining episode provides a great Canary test or documented ground truth against which any quantitative measurea can be compared.

The Antarctic Treaty (The Antarctic Treaty Consultative Parties 1959) entered into force in 1961 and has since grown from 12 original signatories to over 50 consultative and non-consultative parties, making it one of the longest-running successful examples of multilateral environmental governance. Central to its operation is a consensus norm: any Consultative Party can block a decision, creating strong incentives for negotiation and coalition-building (see also Dodds 2010; Joyner 1992).

1.3 Quantitative and computational study of the ATS

The study of strategic interaction among self-interested actors was formalised by Nash (1951) and later extended to evolutionary and repeated-game settings (Axelrod 1984; Osborne and Rubinstein 1994). Consensus dynamics in collective decision-making have been modelled through opinion-pooling and DeGroot-style belief updating (DeGroot 1974), where agents iteratively revise their positions toward a weighted average of their neighbours' views.

1.3.1 Affinity and Aversion Dynamics

1.4 Measuring policy preference from political text

Agent-based modelling (ABM) offers a bottom-up methodology for studying emergent collective behaviour from individual rules (Epstein and Axtell 1996). Applications to political and economic systems have grown substantially (Farmer and Foley 2009), but systematic application to treaty dynamics remains limited.

1.5 Modelling consensus, coalition formation and negotiation

Recent work on AI governance (Dafoe 2018) highlights both the transformative potential and the risks of deploying machine-learning systems in high-stakes institutional contexts. This project uses AI methods (reinforcement learning, network analysis) as *tools* for modelling treaty dynamics rather than as agents within the modelled system.

1.6 Summary of Literature Gap

Chapter 2

Program Design

2.1 Research Questions

Research Questions

RQ1. How do **affinity dynamics** — patterns of positive alignment and coalition-building — shape the emergence and stability of consensus within the Antarctic Treaty System?

RQ2. How do **aversion dynamics** — veto behaviour, strategic defection, and antagonistic positioning — threaten or reconfigure treaty equilibria?

RQ3. What computational simulation framework can reproduce empirically observed ATS decision outcomes, and what generalisable principles does it reveal for multilateral treaty design?

2.2 Theoretical Framework

The project is grounded in three complementary theoretical traditions:

1. **Non-cooperative game theory** — models individual state incentives and deviation conditions (Nash 1951; Osborne and Rubinstein 1994).
2. **Opinion dynamics / consensus models** — captures iterative belief convergence among parties (DeGroot 1974).
3. **Agent-based simulation** — provides a computational substrate for running counterfactual treaty scenarios (Epstein and Axtell 1996).

2.3 Research Design Overview

The project is structured into three interconnected papers, each targeting one research question and one modelling contribution.

Table 2.1: Paper structure and milestones

Paper	Focus	Method	Target venue
1	Affinity dynamics	Network analysis, coalitional GT	TBD
2	Aversion dynamics	Veto-player model, ABM	TBD
3	Simulation framework	Agent-based model, RL	TBD

2.4 Paper 1 — Affinity Dynamics in Multilateral Treaties

2.5 Paper 2 — Aversion Dynamics and Consensus Breakdown

2.6 Paper 3 — Integrated Simulation Framework

2.7 Data Sources and Ethics

The primary data sources for this project are:

- Antarctic Treaty Consultative Meeting (ATCM) final reports and working papers (publicly available at https://www.ats.aq/).
- State-level foreign policy position datasets (e.g., UN voting records).
- Published network and voting data from the international relations literature.

All data are publicly available institutional records; no human-subjects research is involved and no ethics clearance is required beyond standard QUT research integrity protocols.

2.8 Timeline

Table 2.2: Indicative project timeline

Phase	Activity	Target Date
1	Literature review & theoretical framework	Month 6
2	Paper 1: data collection & analysis	Month 12
3	Paper 1: writing & submission	Month 15
4	Paper 2: model development & calibration	Month 18
5	Paper 2: writing & submission	Month 21
6	Paper 3: simulation framework development	Month 27
7	Paper 3: writing & submission	Month 30
8	Thesis integration & write-up	Month 36
9	Thesis submission	Month 38

2.9 Expected Contributions

1. A formal computational model of consensus dynamics in the ATS, the first of its kind at this level of institutional specificity.
2. A transferable agent-based simulation framework applicable to any consensus-based multilateral treaty.
3. New theoretical constructs (affinity/aversion indices) operationalising previously qualitative political science concepts.
4. Policy-relevant insights into treaty design features that improve stability under strategic pressure.

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Appendix A

Supplementary Material

A.1 Notation and Conventions

Table A.1: Notation used throughout the document

Symbol	Meaning
N	Set of treaty parties (agents), $\|N\| = n$
$i, j \in N$	Individual state actors
$s_i \in S$	Strategy of party i
u_i	Utility function of party i
α_{ij}	Affinity score between parties i and j
β_{ij}	Aversion score between parties i and j
$\mathbf{W}$	DeGroot influence/weight matrix
$x_i(t)$	Position/opinion of party i at time t

A.2 Acronyms

ATS Antarctic Treaty System

ATCM Antarctic Treaty Consultative Meeting

ATP Antarctic Treaty Party

ABM Agent-Based Model

RL Reinforcement Learning

GT Game Theory

IR International Relations

QUT Queensland University of Technology

A.3 Candidate's Publication Record

No publications to date; in preparation.